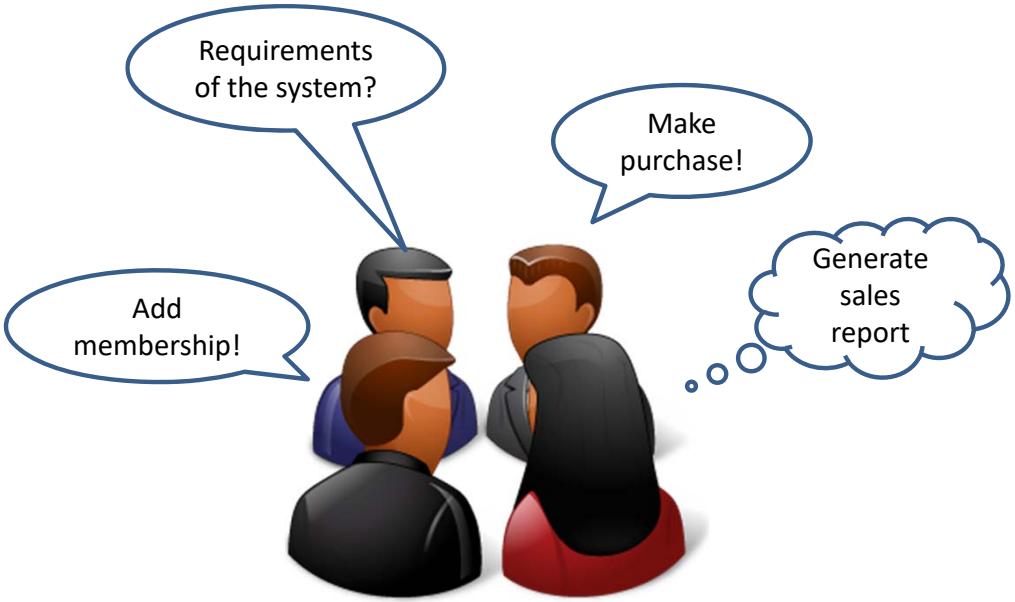
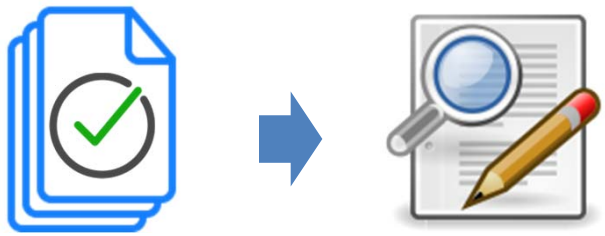
Requirements Analysis

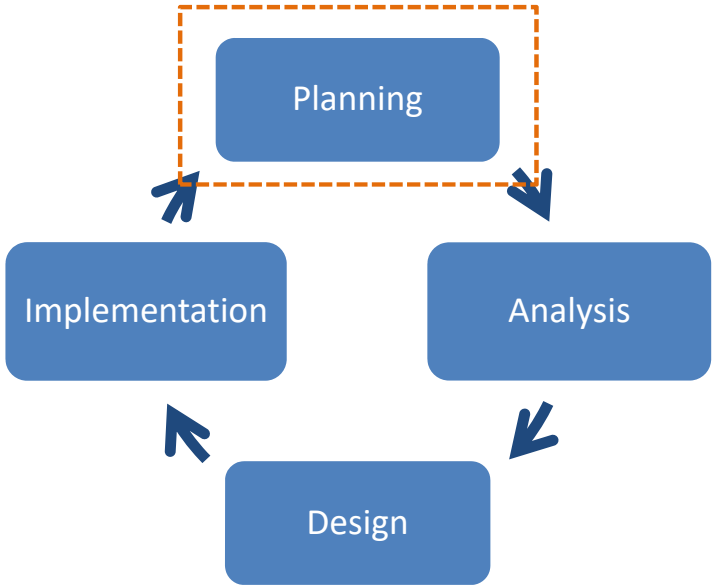


Systems Development Life Cycle

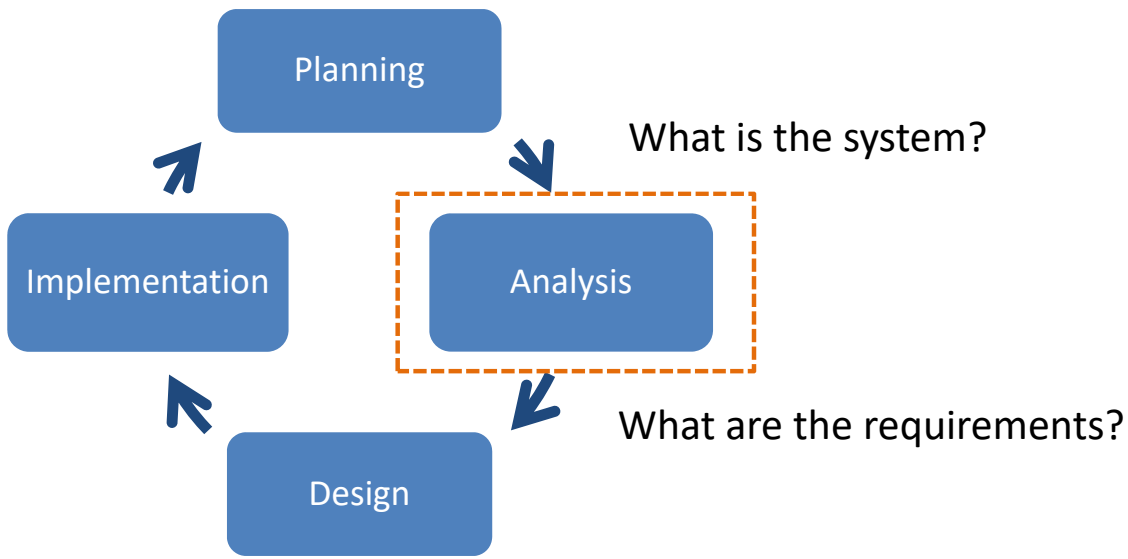


System Request

Feasibility Analysis



Moving to Analysis phase…



Analysis phase is our focus...

Crucial input for <u>design and development</u>

The most challenging phase

The system should be defined properly <u>before</u> we actually build it!

Agenda

Requirements

Functional

Non-functional

Requirements

Functional requirements:

Functions

Business users

Nonfunctional requirements:

Characteristics

System developers

Functional Requirements (1)

"Allow users to achieve objectives"

Functional Requirements

- Allow students to enroll in courses
- Allow professors to view class roster
- Functional requirement 3
- Functional requirement 4
- Functional requirement 5
- ::::::::
- Functional requirement n

Functional Requirements (2)

Functional Requirements of SIS

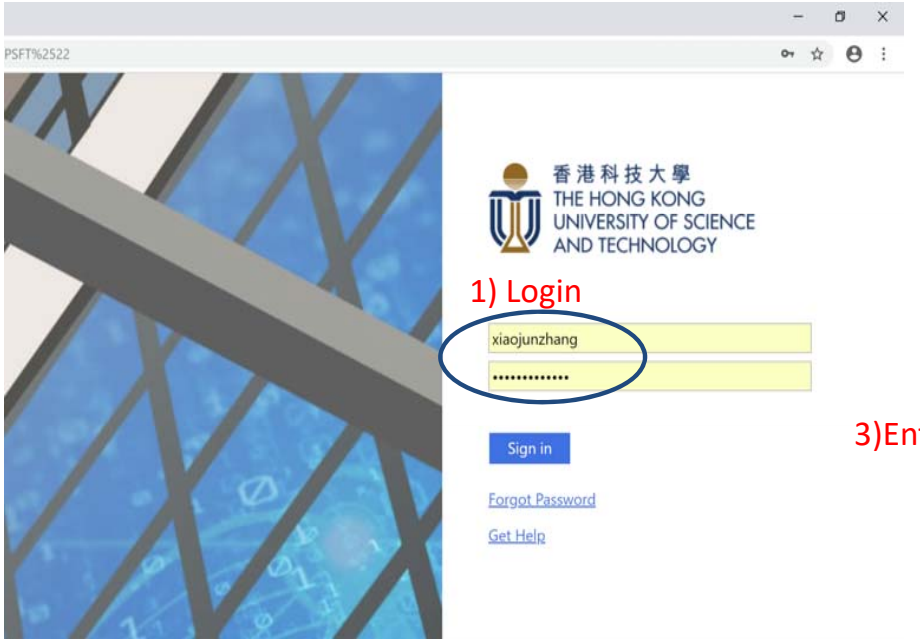
- Allow students to enroll in courses
  - Allow students to search for course information
  - Allow students to register for courses
  - Allow students to cancel courses
- Allow professors to view class roster
- Functional requirement 3
- Functional requirement 4
- Functional requirement 5
- ::::::::
- Functional requirement n

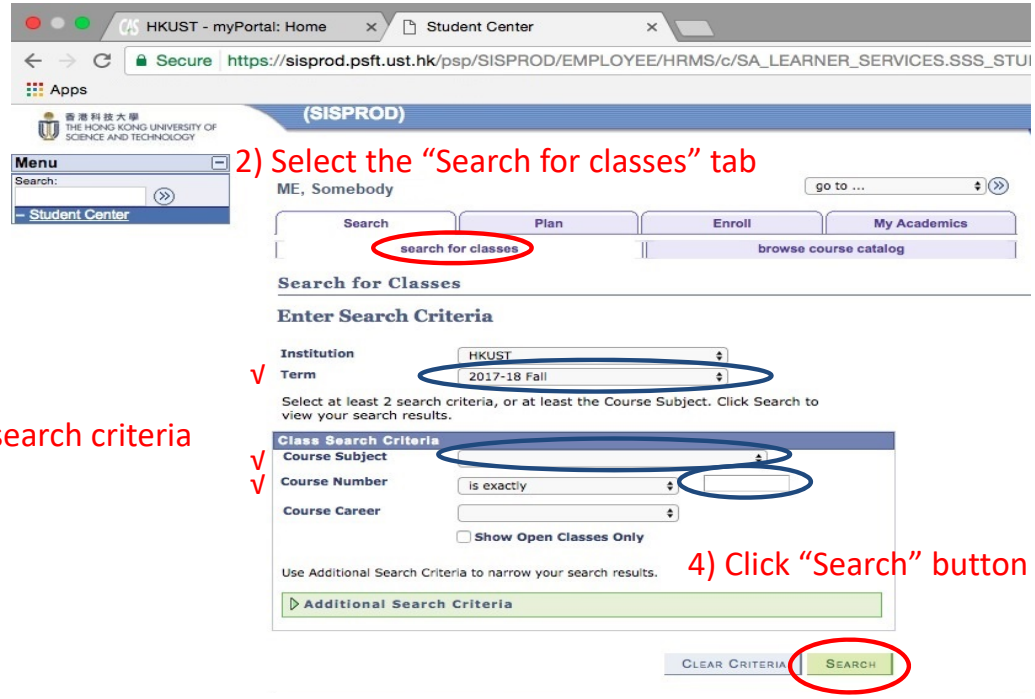
Functional Requirements (3)

Allow students to search for course information

Process

Information





Nonfunctional Requirements (1)

Operational

Security

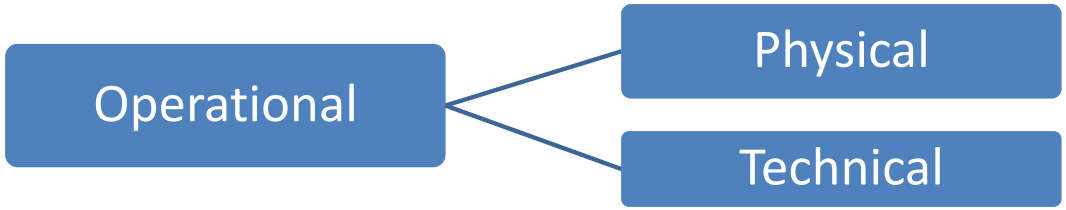
Nonfunctional Requirements

Properties or Characteristics

Performance

Cultural & Political

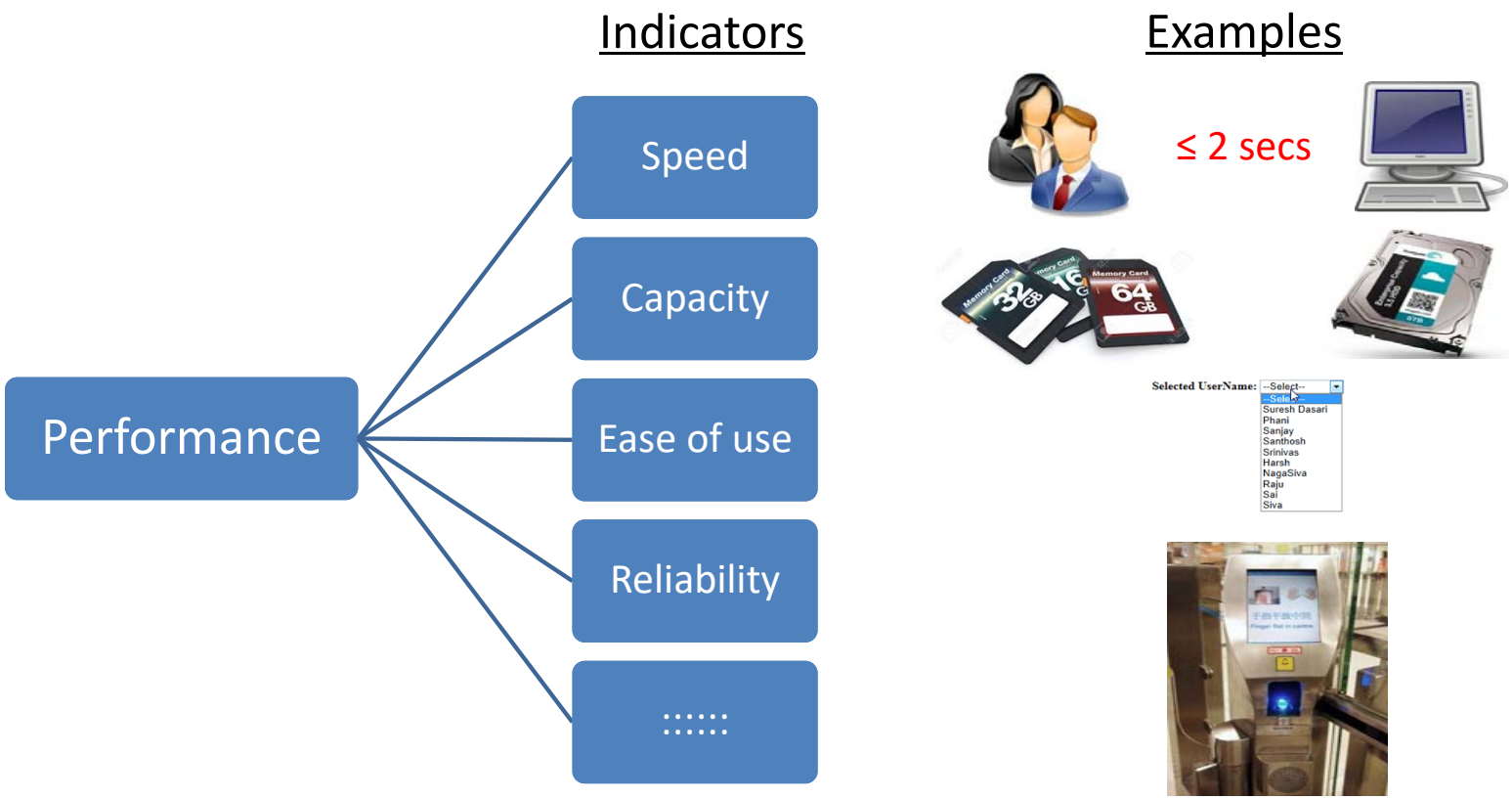
Nonfunctional Requirements (2) - Operational



Physical attributes

Compatibility

Nonfunctional Requirements (3) - Performance



Nonfunctional Requirements (4) - Security

Examples

Security

Private Information

Confidential Information

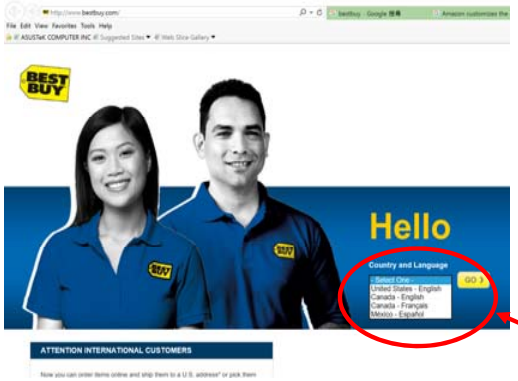
Authorization

Virus Protection

Nonfunctional Requirements (5)- Cultural and Political

Cultural & Political

Examples

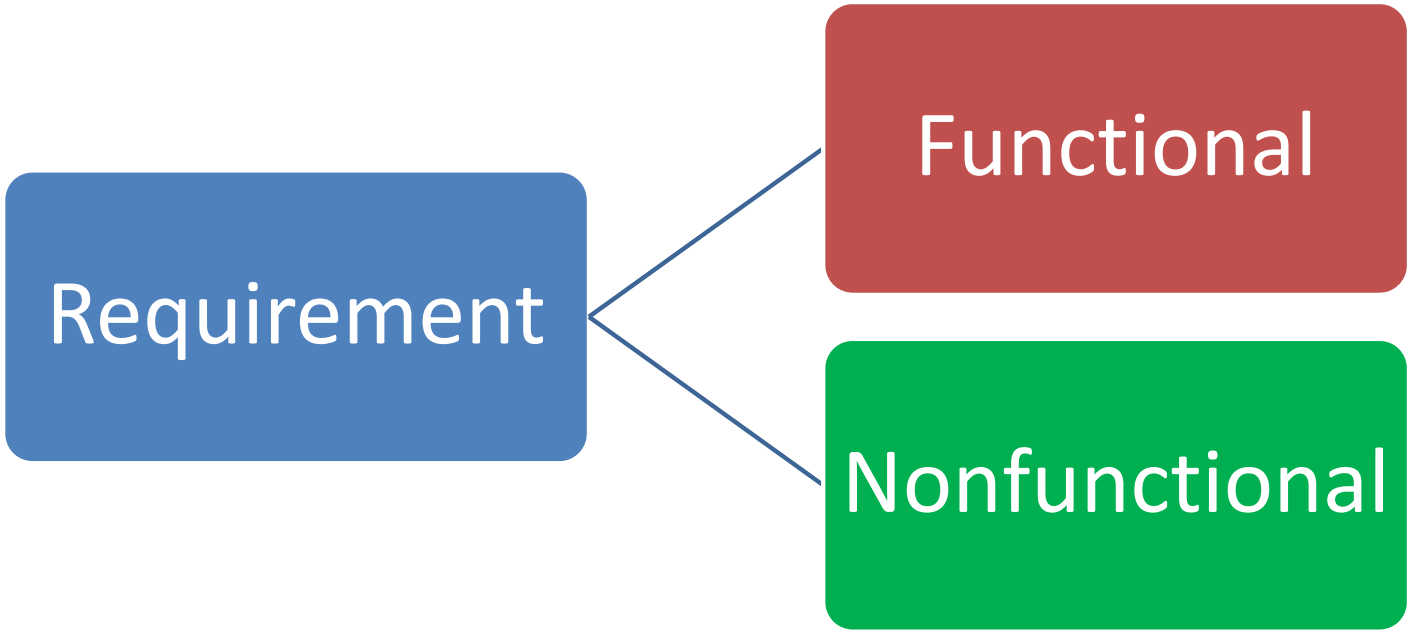


Select country

Medical System

Medical Rules and Standards

Concept Check

- Below is a sentence describes the requirement for a sales system. Is it functional or nonfunctional requirement? If it is a nonfunctional requirement, please specify which dimension of the nonfunctional requirement it is.
  1. Integrate with the accounting system
  2. restrict access to profitability information
  3. provide management reports
  4. have two-second maximum response time for predefined queries and ten-minute maximum response time for ad hoc queries
  5. provide monthly rankings of salesperson performance

Summary